MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

Week 3
Overview of the Earth System

About the Earth System



- All parts of our planet and all of the interactions of those parts comprise the Earth System
- Global geosystems include: climate, plate tectonics, and the geodynamo

The Earth is a dynamic, open system: change is the norm

Source

NASA/Apollo 17 crew; taken by either Harrison Schmitt or Ron Evans - https://web.archive.org/web/20160112123725/http://grin.hq.nasa.gov/ABSTRACTS/GPN-2000-001138.html (image link); see also https://www.nasa.gov/multimedia/imagegallery/image_feature_329.html"The Blue Marble" is a famous photograph of the Earth taken on December 7, 1972, by the crew of the Apollo 17 spacecraft en route to the Moon at a distance of about 29,000 kilometres (18,000 mi). It shows Africa, Antarctica, and the Arabian Peninsula.

Lecture outline

- The scientific method
- Geology as a science

  Geology: from the Greek words for 'Earth' and 'Knowledge'

  - studies all aspects of the planet: its history, composition and internal structure and its surface features
  - addresses practical environmental problems
- Earth's shape and surface
- Discovery of a layered Earth
- Earth as a system of interacting components

Learning objectives

On successful completion of this lecture you will be able to:

- describe key aspects of the scientific discipline of geology that relate it to, and distinguish it from physics, chemistry and biology, and its relevance to the evolution of the Earth and its environment
- illustrate how the scientific method has been used to develop theories of the Earth's shape and structure
- compare the chemical composition and physical characteristics of the Earth's crust, mantle and core

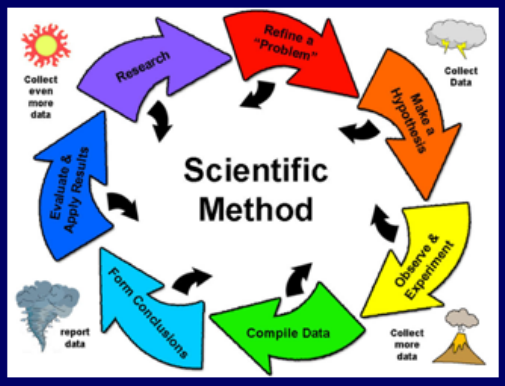
Earth History and the Scientific Method

The goal: explain how the universe works

Observation and experimentation

Development of an hypothesis or multiple hypotheses as tentative explanation(s)

Testing (challenging) and experimentation to eliminate hypotheses or revise them



Scientific theory – a coherent set of hypotheses that explains some aspect of nature

Scientific model – based on many hypotheses and theories

Source
https://geologycafe.com/oceans/images/scientific_method_cycle.jpg

Thought questions for today

How does science differ from religion as a way to understand the world?

If no theory can be proved true, why do almost all scientists believe strongly in General Theory of Relativity or the Standard Model of Big Bang cosmology?

Approaches to Studying the Evolution of the Earth

Major questions in Earth's evolution involve processes that operate on large scales and over long time periods

Geological field observations are supplemented by laboratory experiments

A special aspect of geology: probing Earth's long history by 'reading' the geologic record

They look at mud and see mountains, in mountains oceans, in oceans mountains to be. They go up to some rocks and figure out a story, another rock, another story, and as the stories compile through time they connect- and long case histories are constructed and written from interpreted patterns of clues. This is detective work on a scale unimaginable to moisty detectives, with the notable exception of Sherlock Holmes.

Annals of the Former World, John McPhee



200 million year old desert sand dunes exposed in Colorado, National Monument USA

Source

Mark Newman/Lonely Planet Images/Getty Images, Inc in Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education

The Principle of Uniformitarianism



James Hutton (1726-1797)

The processes that have shaped the Earth through geologic time are the same as those that are observable today



Charles Lyell (1797-1875)

'The present is the key to the past'

Source

Left: https://upload.wikimedia.org/wikipedia/commons/f/f3/Sir_Henry_Raeburn_-_James_Hutton%2C_1726_-_1797._Geologist_-_Google_Art_Project.jpg

Right: https://collectionimages.npg.org.uk/large/mw04054/Sir-Charles-Lyell-1st-Bt.jpg

Process comparison: slow versus fast



Grand Canyon USA

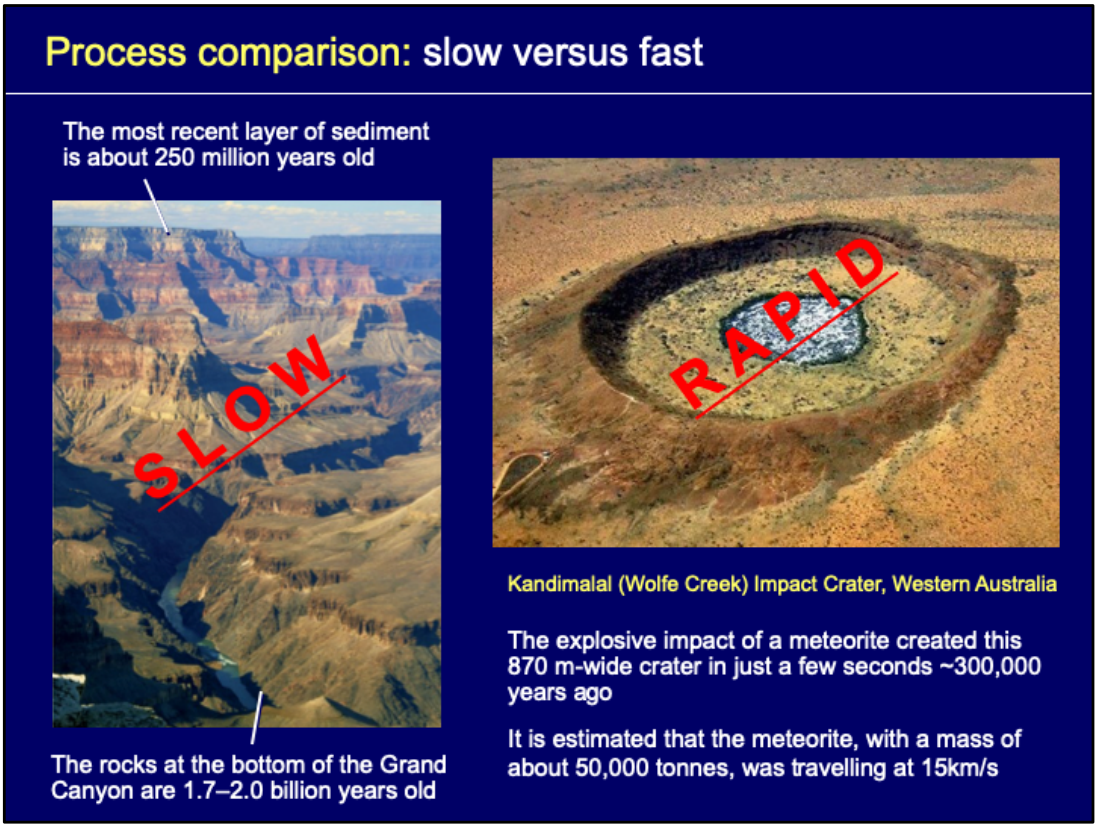


Wolfe Creek Meteor Impact Crater, Australia

Images

Left: John Wang/Photo Disc/Getty Images in in Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education

Right: Wolfe Creek Impact Crator Western Australia (Jamie LaMarche https://www.pinterest.co.uk/pin/466615211377995567/)



Images

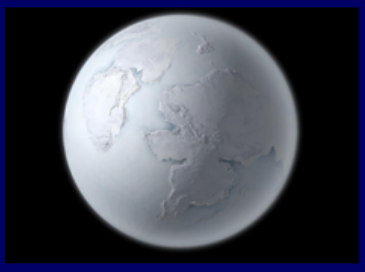
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Right: Wolfe Creek Impact Crator Western Australia (Jamie LaMarche https://www.pinterest.co.uk/pin/466615211377995567/)

The Principle of Uniformitarianism

In strictest, original form: processes act at similar rates to those observed today.

Strict uniformitarianism is untenable because:
- processes occur at different rates through time
- different processes in past which we do not see today
- rare, extreme events on human time scales, are important in Earth history on geologic time scales





To understand that history and how the Earth changed and evolved we need some information about the Earth's shape and interior today

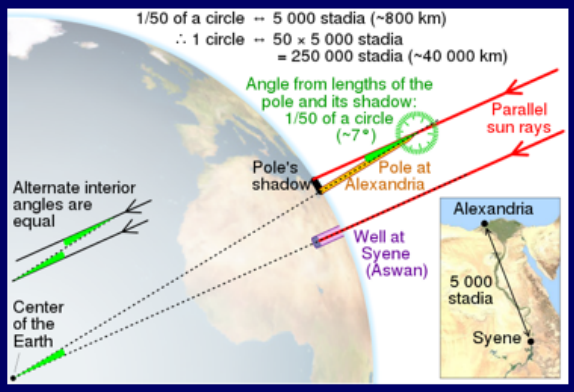
Source

Right: https://images.app.goo.gl/QkbSjBgarrzeNKrS6

Left: https://news.cnrs.fr/sites/default/files/styles/visuel_principal/public/assets/images/cosmos_9035489_001_72dpi.jpg

Earth's Shape and Surface

Geodesy: study of Earth's shape and surface



Earth's centre is ~6,370 km below us ….
The deepest well is 10 km

Modern research shows that Earth is not a perfect sphere

Shape

Ancient Greek scholar Eratosthenes (*c.* 276 - *c.* 194BC)

- measured the circumference and radius of the Earth
- hypothesized that Earth was spherical



Source

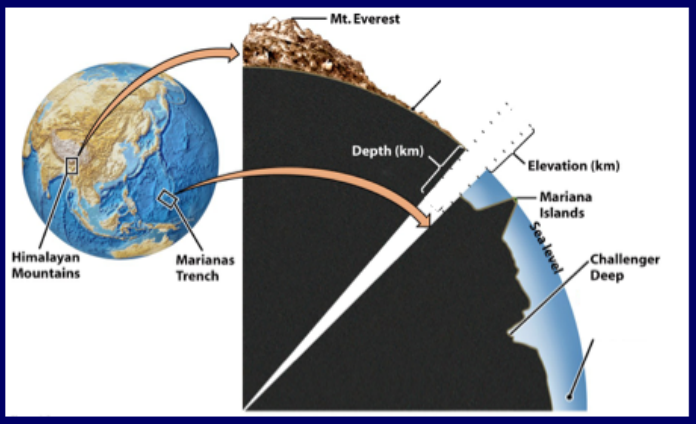
https://upload.wikimedia.org/wikipedia/commons/thumb/1/1f/Eratosthenes_measure_of_Earth_circumference.svg/640px-Eratosthenes_measure_of_Earth_circumference.svg.png

Right

https://svs.gsfc.nasa.gov/vis/a010000/a011000/a011010/space_geodesy_prores_03341_1280x720_ipad_1024x576.jpg

Earth's Shape and Surface

Geodesy: study of Earth's shape and surface



Surface Elevation

- Sea level – reference level for all Earth surface and seafloor elevations

- Earth's topography (surface elevation) ranges nearly 20 km from highest to lowest point
  - Mt Everest: 8850 m above sea level
  - Challenger Deep: 11,030 m below sea level

The interior of the Earth has layers

Source

Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education

How do we Know the Earth has Layers?

Surface rock density <3.5 g/cm^3
- measure mass/volume

But, whole Earth density is 5.5 g/cm^3
1798 Henry Cavendish
- volume (Eratosthenes)
- mass (Newton's law of gravity- strength of gravitational pull depends on mass present)



Granite density 2.7 g/cm^3

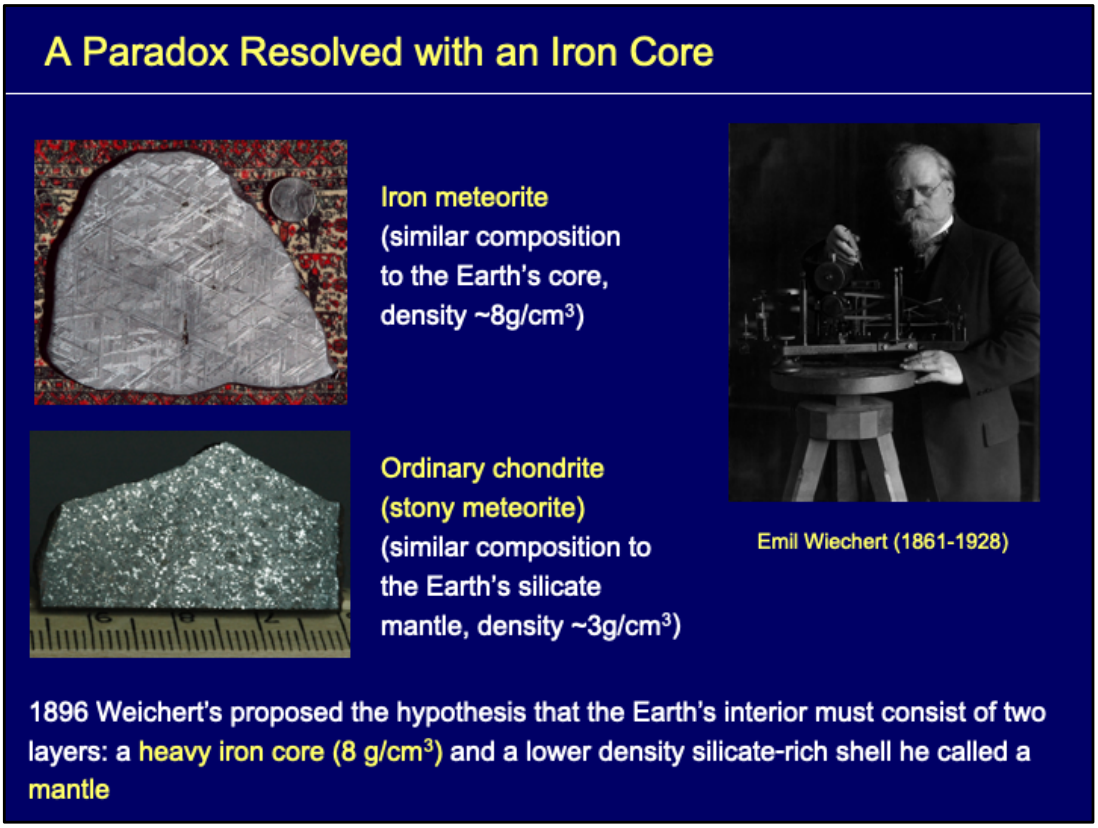


Mantle rocks brought to the Earth's surface by volcanoes (Olivine: iron-magnesium silicate density 3.5 g/cm^3)

Source

Top: A tombstone made from Granite. Granite is favoured over limestone due the greater resistance of the constituent minerals to chemical weathering,

Bottom: East Ridge of Mt Terror, Antarctica. Mantle xenolith represent the deep mantle below the crust (20-100kms deep). Green crystals are olivine and black crystals are pyroxene (https://blog.nationalgeographic.org/2012/11/12/xenoliths-from-the-mantle-little-green-rocks-from-deep-inside-the-earth/)



Source

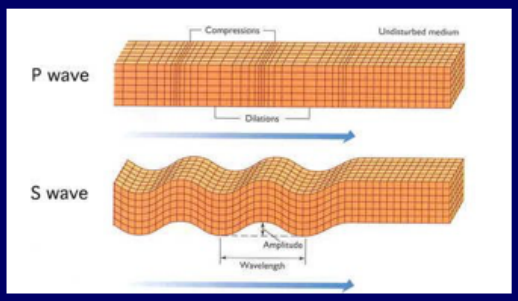
Left top: Sawn, polished and etched slab of the Gibeon iron meteorite. This meterite has a distinct Widmanstatten pattern. Photo Randy Korotev. (http://meteorites.wustl.edu/id/metal.htm)

Left bottom: Metal grains reflecting light in a polished slice of the ordinary chondrite Faucett (H5). Photo Randy Korotev. (http://meteorites.wustl.edu/id/metal.htm)

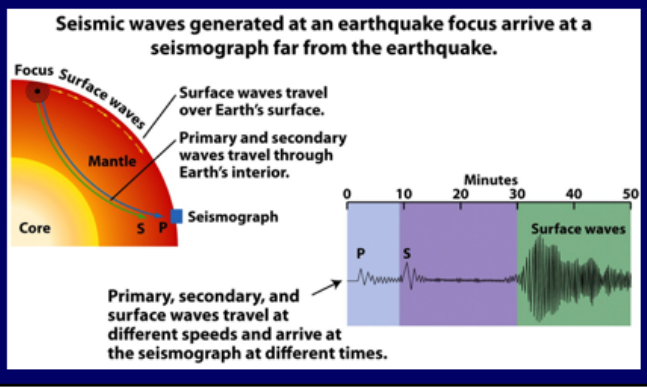
Right: Undated picture of Emil Weichert (taken by A. Schmidt) property of the Institute of Geophysics, Georg August University Göttingen

Seismic Waves Illuminate Earth's Interior

Seismic waves: waves produced by earthquakes
Body waves: pass through the interior of the Earth compressional (**P waves**) and shear waves (**S waves**)
Surface waves: travel along the Earth's surface



Seismic waves, recorded on seismometers allow us to locate earthquakes and make pictures of the interior of the Earth



Source

Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education

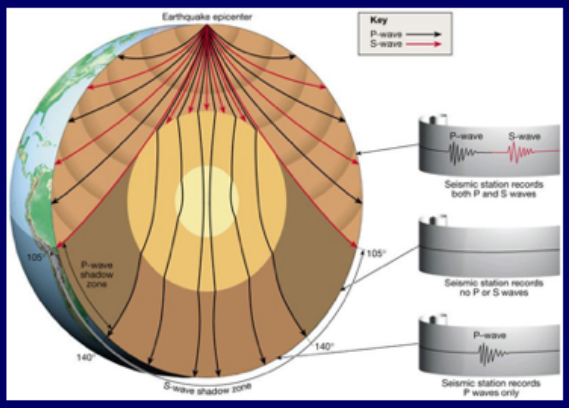
Seismic Waves Illuminate Earth's Interior

P and S waves behave differently

- P waves: travel trough solid, liquid and gas (~6 km/s)
- S waves: travel through solids only (~3 km/s)

P and S waves are bent (refracted or reflected) or absorbed at layer boundaries within Earth

- seismic waves travel at different velocities through different materials

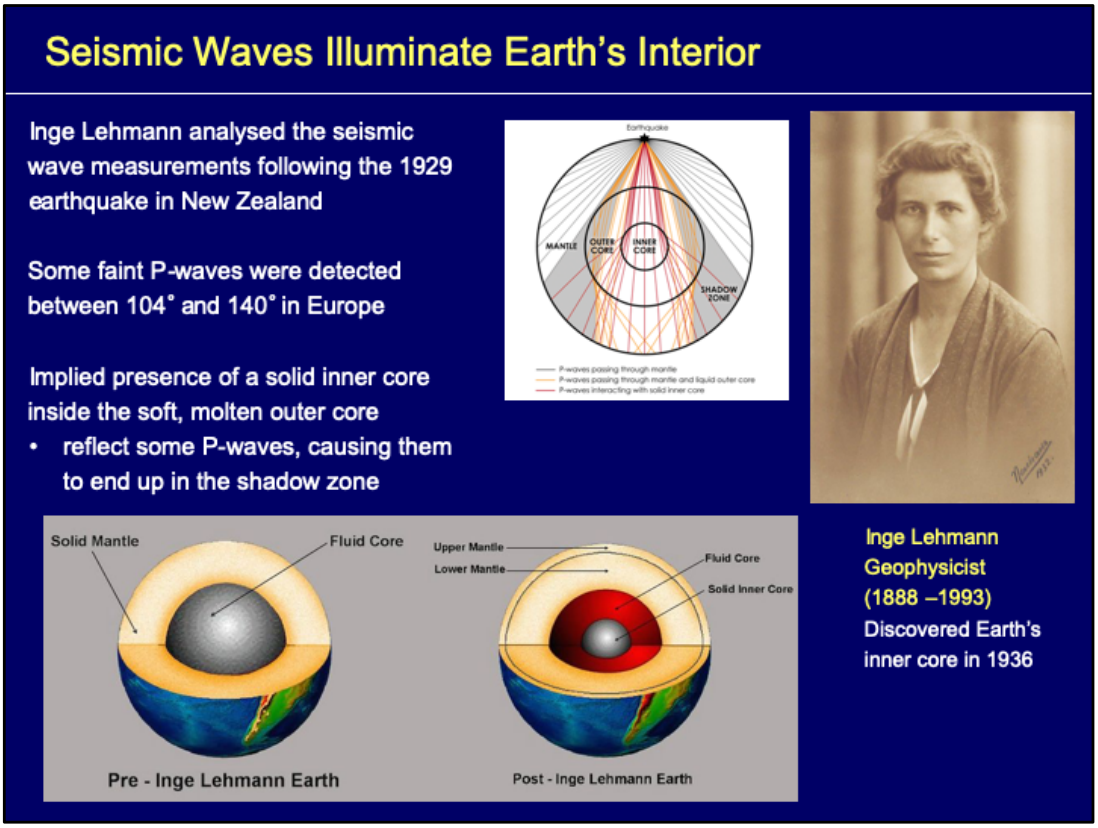


Observations implied:

- the core is a molten, semi-liquid

Source

https://www.vox.com/2015/5/13/8595157/inge-lehmann



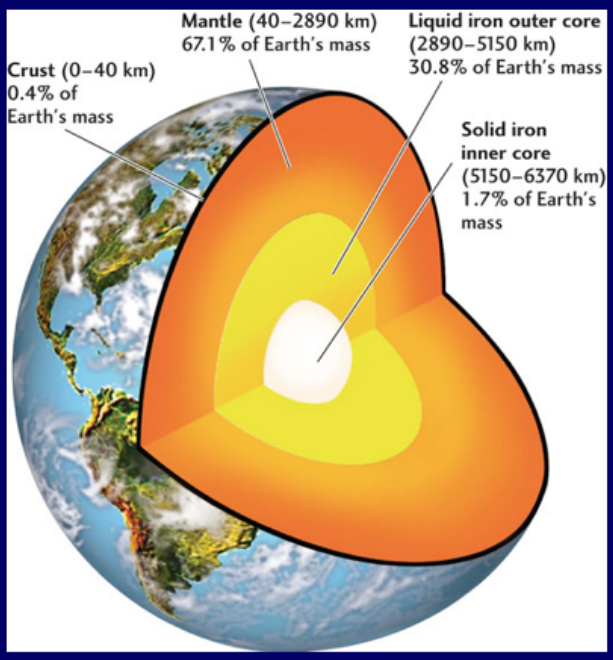
Source

Top left: https://www.earthmagazine.org/article/march-29-1936-notes-earths-inner-core

Top right: Inge Lehmann, shown in 1932. Image courtesy The Royal Library, National Library of Denmark, and University of Copenhagen University Library

Bottom: The structure of the Earth https://www.vox.com/2015/5/13/8595157/inge-lehmann

Earth's Internal Structure

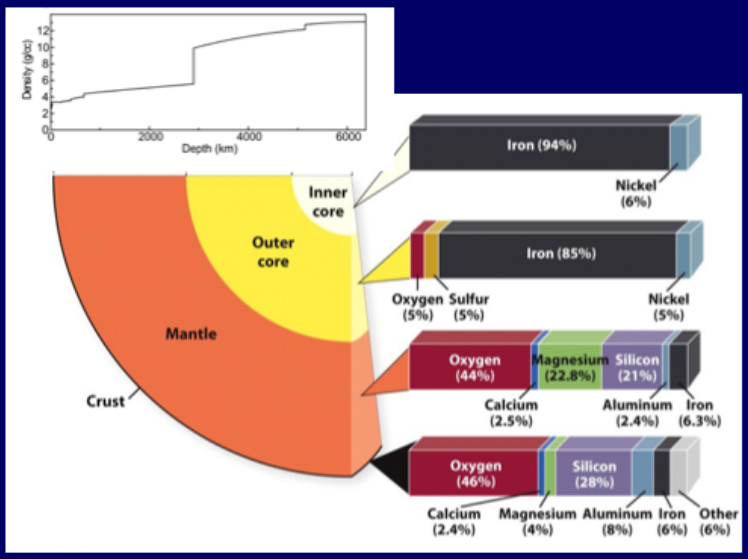


Source

Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education

Composition of a Layered Earth

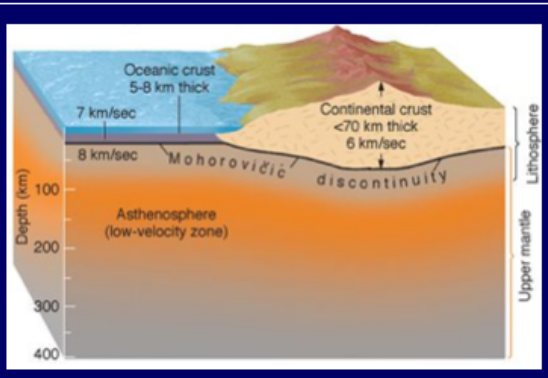
Abrupt changes in density between Earth's major interior layers are caused by changes in the chemical composition of those layers



Source

Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education

Earth's Crust

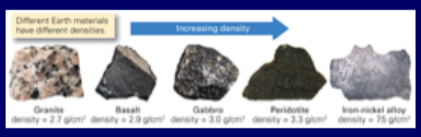


Outer layer of rock

- forms thin skin on Earth's surface (0.1-1.0% of Earth's radius)

The crust-mantle boundary is marked by a jump in seismic wave velocity

= Mohorovicic discontinuity or Moho



Two fundamentally different types of crust:

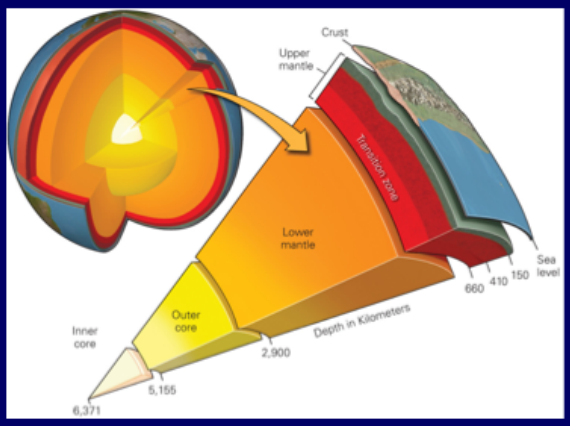
Oceanic crust:
- underlies sea floor
- thinner and denser
- av. thickness 7 km
- seismic wave velocities ~7 km/s
- composition mafic: basalt and gabbro
- result of mantle melting, deeper levels

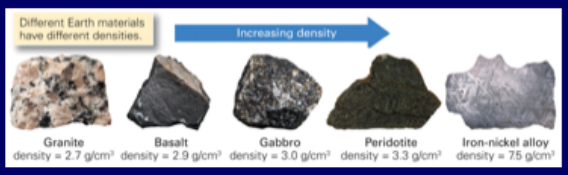
Continental crust:
- underlies continents
- thicker and less dense
- av. thickness 40 km
- seismic wave velocities ~6 km/s
- composition more felsic: granite-like
- derived by a sequence of melting processes

Source

McGrawiHill

Earth's Mantle





crust mantle core

- 2820-2890 km thick shell that surrounds the core
- 67% of Earth's mass
- Composition: higher density peridotite
- Solid rock (with tiny drops of molten rock at depths of 100-300 km)
- Undergoes slow convection

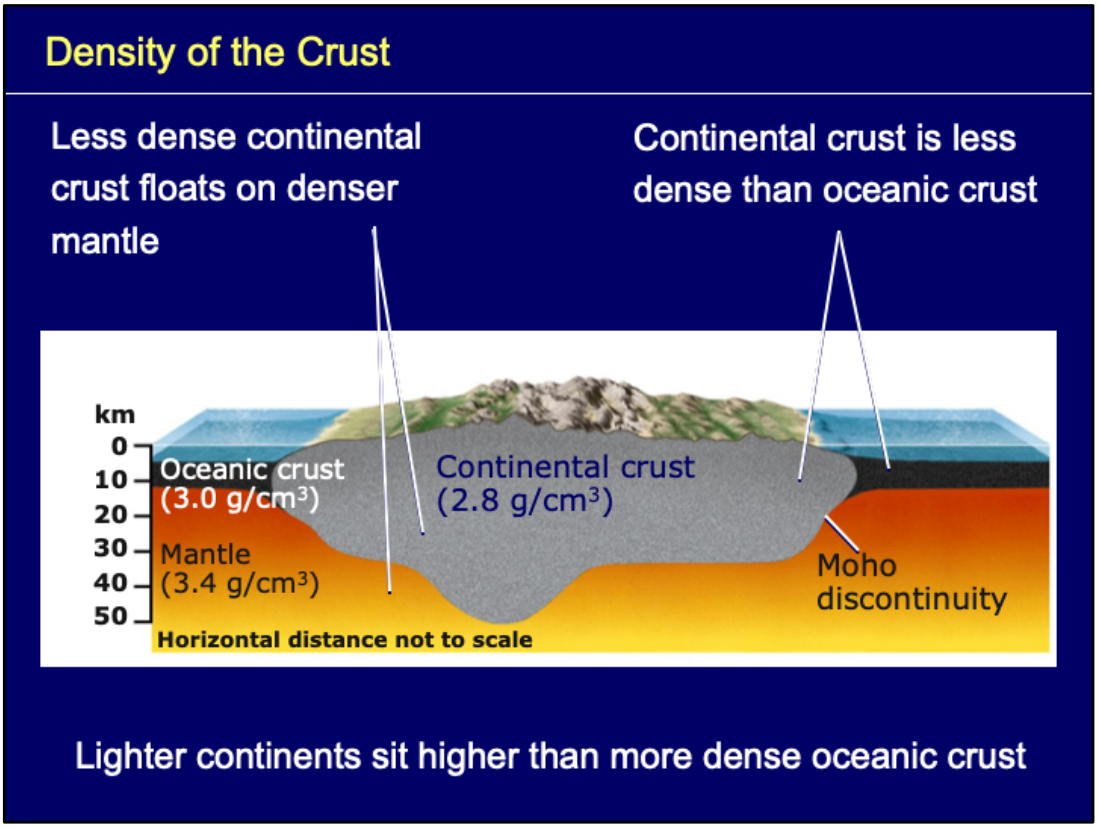
Step-wise changes in velocity of seismic waves define sublayers

- 410km Olivine-spinel density change, not compositional
- 660km boundary to lower mantle

Boundary with core where mantle plumes arise from

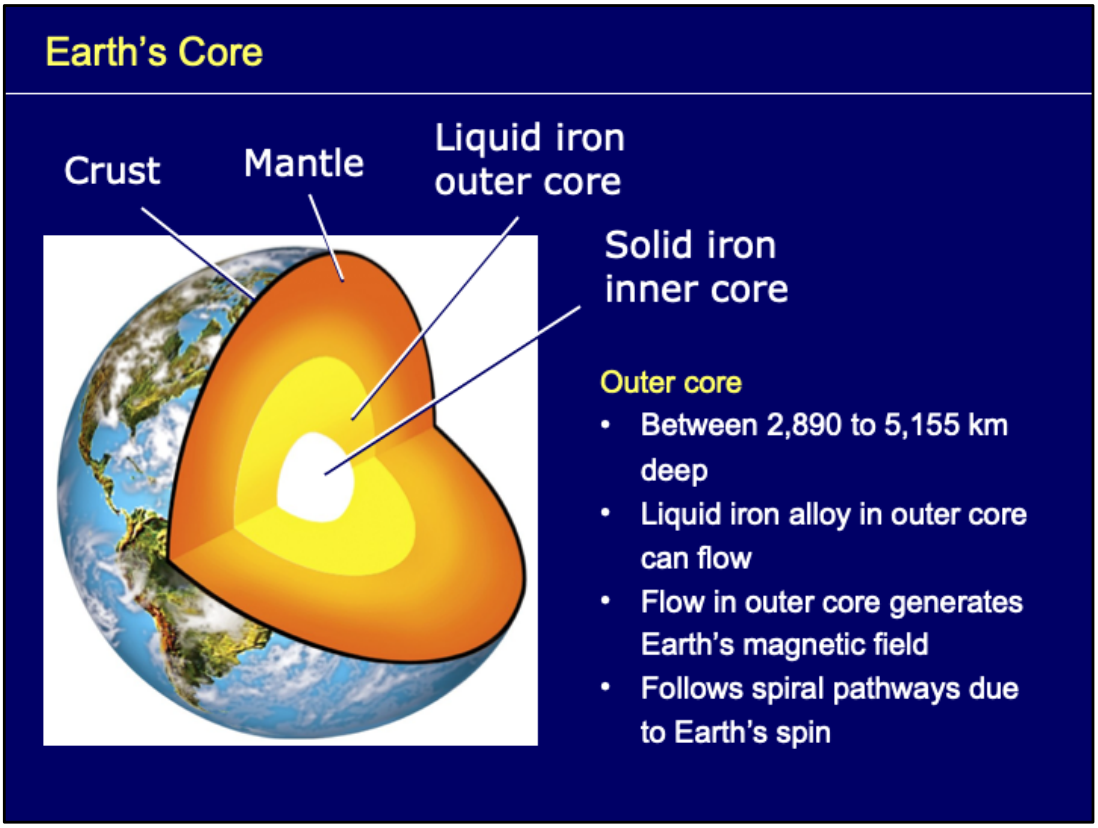
Source

Marshak, 2019, Earth Portrait of a Planet



Source

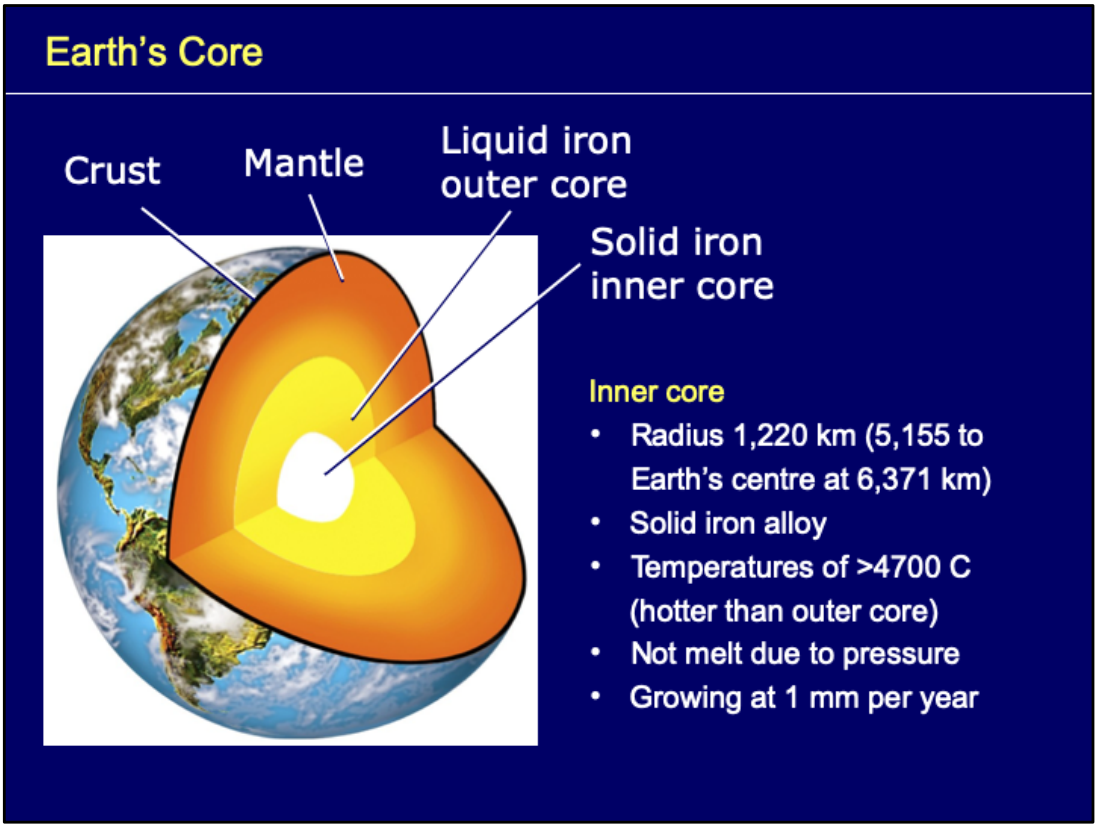
Grotzinger and Jordan, 2014, Understanding Earth, Macmillan Education



Studies of seismic waves have subdivided the core into two parts, the outer core and the inner core.

Image

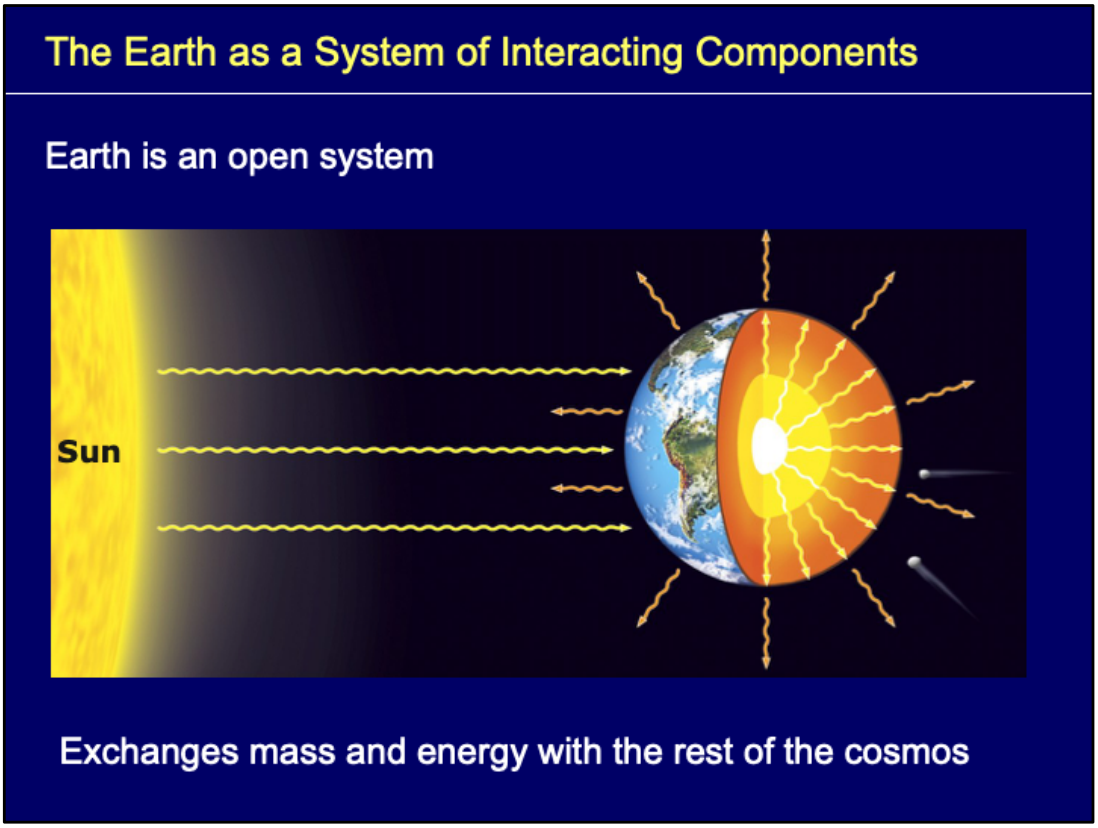
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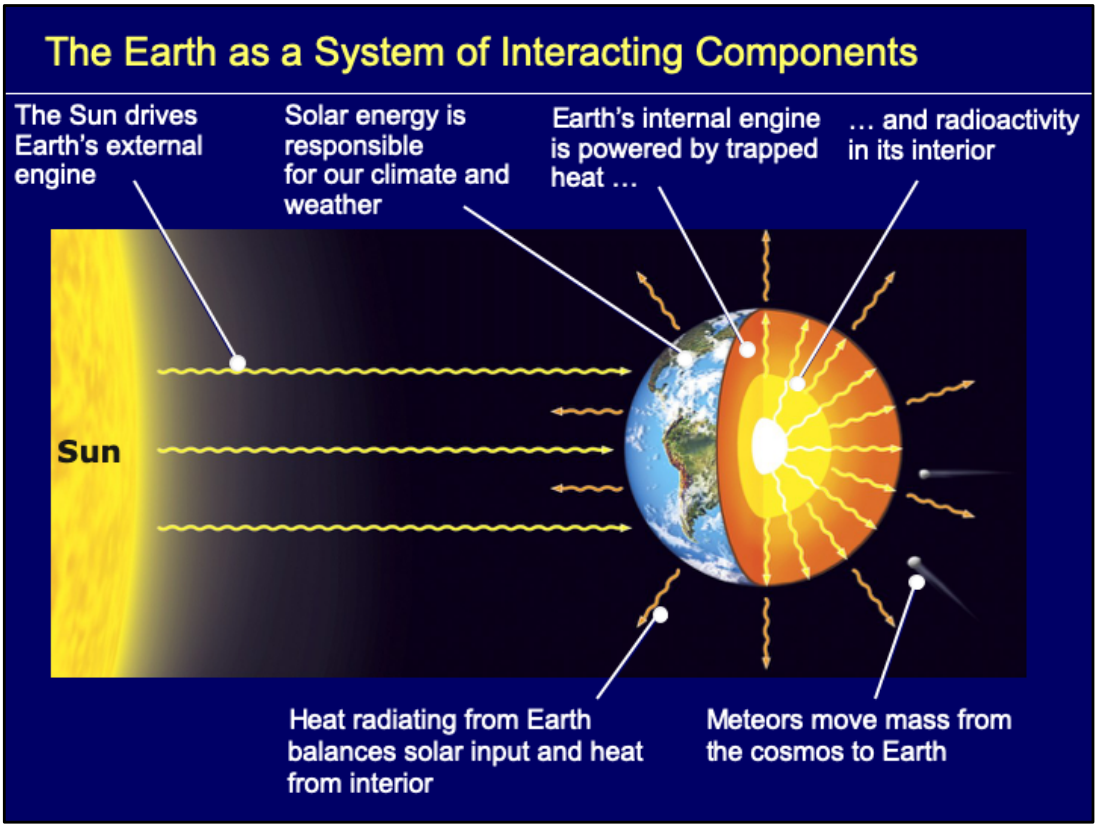
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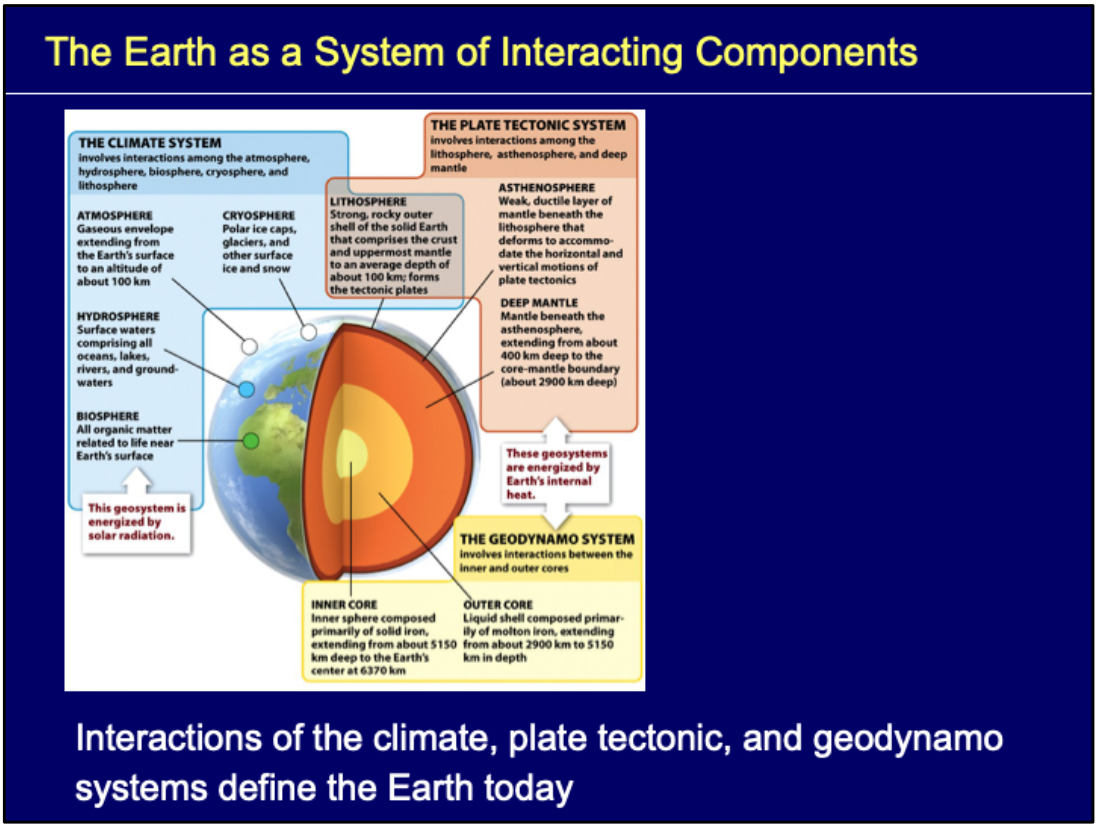
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